

AMULYA B

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Professional Summary

AI & ML graduate with hands-on experience building AI-powered healthcare, mobility, and full-stack web applications using Python, Flask, FastAPI, and Machine Learning. Developed projects involving IoT monitoring, intelligent ride allocation, and real-time analytics systems. Completed a Python Full Stack Development Internship at Dhee Coding Lab with exposure to backend integration, REST APIs, and deployment workflows.

Technical Skills

- **Programming Languages:** Python, JavaScript, SQL
- **Web Technologies:** HTML, CSS, React
- **Frameworks & Libraries:** Streamlit, Flask, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow (basic), OpenCV
- **Database & Backend:** SQL, REST APIs, CRUD Operations
- **Tools & Platforms:** Git, GitHub, Google Colab, Power BI, Excel
- **Areas of Interest:** AI-powered applications, mobility platforms, healthcare technology

Projects

Sleep Apnea Detection using ML & IoT GitHub | Live Demo

- Built an IoT-based sleep apnea monitoring system using ESP32-C3 and MAX30102 sensors for heart rate and SpO₂ tracking.
- Developed a Random Forest ML pipeline with Firebase cloud storage and Streamlit dashboard for real-time monitoring and alerts.
- Processed real-time physiological sensor data for desaturation event analysis and prediction.

SafeHer Ride — AI-Powered Women Safety Mobility Platform GitHub | Live Demo

- Engineered a full-stack women safety ride-booking platform using Python and Streamlit with modular backend architecture.
- Implemented AI-driven driver matching using Dijkstra's shortest path algorithm, safety-score prioritization, and route optimization for secure ride allocation.
- Developed booking workflows, emergency SOS systems, verified driver management, live route visualization, and ride lifecycle control.
- Built interactive rider and driver dashboards with AI-assisted safety insights and route intelligence features.

Habit Tracker — Full Stack Web Application GitHub | Live Demo

- Built and deployed a full-stack habit tracking application using React and FastAPI with RESTful APIs.
- Implemented CRUD operations, API versioning, and Swagger/OpenAPI documentation.
- Deployed frontend and backend services using Vercel and Railway.

Internship Experience

Python Full Stack Development Intern — Dhee Coding Lab Present

- Working on Python full-stack application development with backend integration and frontend workflows.
- Gaining practical experience in modular application architecture, debugging, deployment workflows, and REST-based systems.
- Collaborating on real-world web application development and project implementation.

Education

PES Institute of Technology and Management, Shivamogga 2022 – 2026

Bachelor of Engineering in Artificial Intelligence and Machine Learning
CGPA: 8.0

Paper Presentation

Sleep Apnea Detection using Machine Learning and IoT

Presented at the *2nd International Symposium on Collaborative Informatics (ISCI-2025)*, University BDT College of Engineering, Davangere, Karnataka Dec 2025

Positions of Responsibility

• IEEE Student Secretary, PESITM	2024 – 2026
• IEEE Volunteer and Student Member, PESITM	2022 – 2024
• Finance Coordinator, AIML Department	2024 – 2025
• Forum Coordinator	2024 – 2026
• Cultural Coordinator — Organized and coordinated student events and activities	2023 – 2025

Certifications

• Data Analysis Using Python — IBM
• Deep Learning with TensorFlow
• Attention in Transformers (PyTorch)

Activities & Achievements

• Smart India Hackathon — RainWise Project
• Cybersecurity Bootcamp on 6G — NITK Surathkal
• District-Level Swimming Championship Participant

Interests

• Reading
• Dancing
• Public speaking and communication